

Roll 4358 · Maryam Sana

Neonatal ICU Vital Signs Automation System

Advanced

PROJECT DETAILS

Premature Infant Monitor Firmware Simulator

Simulates embedded firmware of a NICU incubator. IncubatorUnit class controls temperature, humidity, and oxygen concentration. VitalMonitor tracks heart rate, SpO2, and respiration rate for each infant. AlarmManager uses a tiered alert system (Advisory / Warning / Crisis). NurseStation class manages multi-bed triage queue sorted by severity. PhysicianLog records interventions with timestamps. Mirrors GE Carescape / Philips IntelliVue NICU system architecture.

Graphics Output

Use SFML to render a multi-bed NICU dashboard — each bed shown as a panel with scrolling SpO2 and heart rate waveforms. Incubator temperature displayed as a color-gradient dial. Crisis alerts flash the entire bed panel red with an audible-style animation. Triage queue on the right reorders beds by severity in real time.

SFML multi-panel

Healthcare IoT Triage Logic Firmware Sim

Roll 4523 · Saba

Nuclear Reactor Core Safety Monitor

Advanced

PROJECT DETAILS

Reactor Protection System (RPS) Firmware Sim

Simulates a nuclear reactor protection system. ReactorCore class models neutron flux, coolant temperature, pressure, and control rod positions. SafetyChannel classes (4 redundant channels) independently vote on trip conditions using 2-of-4 logic — exactly as in real reactor designs. ScramController initiates emergency shutdown when voted. EventLog records all safety events with microsecond timestamps. Models Three Mile Island / PWR reactor instrumentation and control logic.

Graphics Output

Use SFML to draw a reactor schematic — core shown as a central rectangle with animated heat shimmer effect, coolant loop as animated flow lines, and 4 safety channel indicators as colored status lights. When a SCRAM is triggered the entire screen flashes red with a "REACTOR TRIP" banner and control rods animate

Roll 4359 · Eman Fatima

Neural Signal Brain-Computer Interface Simulator

Advanced

PROJECT DETAILS

EEG-Based Thought-to-Command Decoder

Simulates a Brain-Computer Interface (BCI) system. EEGChannel class generates synthetic brainwave signals across delta, theta, alpha, beta, and gamma bands using superimposed sine waves. BandpowerExtractor computes power in each frequency band via sliding window. IntentClassifier maps power patterns to commands (left/right/select/rest) using threshold logic. CommandDispatcher sends decoded commands to a virtual device. Models real OpenBCI / Emotiv hardware pipeline entirely in software.

Graphics Output

Use SFML to render a scrolling multi-channel EEG strip (5 frequency bands as stacked waveforms in different colors). A brain topography circle diagram shows which regions are active. Decoded command flashes in large text at the bottom (MOVE LEFT / SELECT). Band power shown as a live horizontal bar chart updating each window. Looks exactly like BCI research software.

SFML waveform strips

BCI / Neuroscience

Signal Processing

Pattern Classify

Roll 4556 · Zarmeen

Edge AI Inference Engine for IoT Devices

Advanced

PROJECT DETAILS

TinyML Runtime Simulator for Microcontrollers

Simulates TinyML inference as it runs on microcontrollers. NeuralLayer class hierarchy (DenseLayer, ReLULayer, SoftmaxLayer) implements forward pass math. ModelLoader reads a simple JSON-format model definition. InferenceEngine runs sensor data through the model and outputs class predictions. MemoryTracker enforces strict RAM/Flash budgets matching real MCU constraints (e.g. 256KB). LatencyProfiler measures per-layer execution time. Models TensorFlow Lite Micro / Edge Impulse runtime behavior.

Graphics Output

Use SFML to visualize the neural network as animated layer boxes — neurons as circles, activations shown as brightness intensity. During inference, a data pulse animates through each layer left to right. Output probabilities render as a live bar chart. Memory

downward. Coolant temperature gauge needle sweeps into the danger zone before trip. Extremely dramatic and realistic.

SFML schematic

Safety Systems Nuclear I&C Voting Logic

Roll 4559 · Abeeha Advanced
Software-Defined Radio (SDR) Signal Decoder

PROJECT DETAILS

RF Signal Processing & Protocol Decoder Engine

Simulates a software-defined radio receiver pipeline. RFSource generates synthetic IQ samples for AM, FM, and ASK-modulated signals. Demodulator class hierarchy handles each modulation type polymorphically. FrameDecoder extracts data frames from the demodulated bitstream using preamble detection and CRC checking. SpectrumAnalyzer computes FFT magnitude from IQ samples. Models the actual RTL-SDR / GNU Radio processing pipeline in pure C++.

Graphics Output

Use SFML to render a 3-panel SDR display: top panel shows a live spectrum waterfall (frequency vs. time, color-mapped by signal strength — exactly like SDR# software), middle panel shows the demodulated waveform, bottom panel shows decoded frame bytes in hex. Tuned frequency marker is a draggable vertical line. This is the most technically impressive visual in the entire course.

SFML waterfall display

SDR / RF Signal Decoding FFT Spectrum

Roll 4561 · Muskan Ali High
Smart Dialysis Machine Controller

PROJECT DETAILS

Hemodialysis Session Management Firmware

Simulates the control firmware of a hemodialysis machine. BloodPump and DialysatePump classes model flow rates (mL/min). MembraneFilter class simulates toxin clearance using urea kinetic modeling. PressureMonitor detects venous/arterial pressure alarms. HeparinPump class delivers anticoagulant doses on schedule. SessionManager tracks treatment time, Kt/V clearance target, and fluid removal goals. Mirrors Fresenius 5008 / Baxter hemodialysis machine firmware architecture.

Graphics Output

Project Topics.html

usage bar fills up as model loads. Latency per layer shown as a timeline strip below the network diagram.

SFML neural viz

TinyML / Edge AI OOP Layer Design MCU Simulation

Roll 4560 · Amna Shoukat Advanced
CAN Bus Automotive ECU Network Simulator

PROJECT DETAILS

Vehicle Electronic Control Unit Communication Engine

Simulates a car's CAN bus network connecting multiple ECU nodes: EngineECU, ABS_ECU, AirbagECU, DashboardECU, and TransmissionECU. Each ECU broadcasts CAN frames with arbitration IDs and data payloads. CANBus class manages message arbitration using priority-based ID resolution. DiagnosticTool class implements OBD-II PID queries (engine RPM, coolant temp, vehicle speed). FaultCodeManager stores and clears DTCs. Models real automotive network protocols used in every modern vehicle.

Graphics Output

Use SFML to render a car dashboard HUD — speedometer and RPM gauge as animated needles, engine temperature and fuel level as bar gauges. A CAN bus traffic panel shows live scrolling frames with arbitration IDs color-coded by source ECU. When a DTC fault fires, a warning icon lights up on the dashboard and the relevant ECU node flashes in the network diagram. Looks like a real OBD-II diagnostic tool screen.

SFML dashboard HUD

CAN Bus Protocol Automotive IoT OBD-II / DTC

Roll 4562 · Laiqa Fatima Advanced
Distributed RAID Storage Controller Simulator

PROJECT DETAILS

Enterprise Storage Array with Fault Tolerance Engine

Simulates a hardware RAID controller managing an array of disk drive objects. DiskDrive class models read/write latency, failure probability, and sector map. RAIDController implements RAID-0, RAID-1, RAID-5, and RAID-6 striping/parity logic. RebuildEngine reconstructs lost data from parity when a drive fails. IOScheduler queues and dispatches read/write requests using elevator algorithm. HealthMonitor tracks SMART attributes per drive. Models LSI MegaRAID / HPE Smart Array controller behavior.

Use `winbgi` to render the dialysis circuit — blood flow path as red animated lines, dialysate as blue lines, filter as a central rectangle. Flow rate gauges update in real time. A session progress bar shows Kt/V clearance rising toward target. Pressure alarms flash the relevant sensor node. Fluid removal counter counts down. Looks exactly like the touchscreen on a real dialysis machine.

`winbgi` circuit diagram

Medical Device IoT Urea Kinetics Firmware OOP

Roll 4563 · Laiba Javed High
Seismic Early Warning & Earthquake Monitor

PROJECT DETAILS
Seismograph Network & P-Wave Detection Engine
Simulates a national seismograph network. `SeismicSensor` class generates synthetic ground motion waveforms with P-wave and S-wave arrivals. `WaveDetector` uses STA/LTA (Short-Term Average / Long-Term Average) algorithm — the real method used by USGS — to detect event onset. `TriangulationEngine` calculates earthquake epicenter from 3+ station arrival times. `MagnitudeCalculator` applies Richter scale formula. `AlertBroadcaster` issues warnings ranked by estimated impact zone. Models Pakistan Meteorological Department seismic network logic.

Graphics Output
Use `SFML` to show a map of Pakistan with seismograph stations as triangle icons. When a quake is detected, animated concentric circles expand from the epicenter. Each station's waveform scrolls in a mini-panel. STA/LTA ratio plotted as a secondary trace — crossing the threshold triggers a red detection marker. Magnitude and location display updates in real time. Dramatically realistic for a disaster management demo.

`SFML` map + waveforms

STA/LTA Algorithm Disaster IoT Triangulation

Roll 4565 · Sana Sidique Advanced
Underwater ROV Navigation & Sonar System

PROJECT DETAILS
Remotely Operated Vehicle Control & Mapping Sim
Simulates an underwater ROV (Remotely Operated Vehicle) used for pipeline inspection and ocean floor mapping. `ROVController` manages thruster groups (surge, heave, sway, yaw) as

Graphics Output
Use `SFML` to render the disk array as a row of drive bay icons — green for healthy, amber for degraded, red for failed. When a drive fails, data rebuild animates as a progress bar filling the replacement drive. Stripe layout visualized as a colored grid showing data/parity blocks across drives. IO activity shown as flashing read/write indicators per drive. RAID level selector changes the stripe pattern live.

`SFML` drive array viz

Storage Systems RAID Parity Logic IO Scheduling

Roll 4564 · Areeba Kanwal Advanced
Machine Vision Quality Control Inspector

PROJECT DETAILS
Inline PCB Defect Detection & AOI System Sim
Simulates an Automated Optical Inspection (AOI) system for PCB manufacturing. `PCBBoard` class stores a grid of component slots with attributes (position, size, orientation, solder quality). `CameraCapture` generates synthetic inspection images as 2D arrays. `DefectDetector` class runs a rule-based inspection pipeline: missing components, misaligned ICs, solder bridging, and tombstoning. `PassFailJudge` applies accept/reject thresholds. `ConveyorController` paces boards through inspection zones. Mirrors Cognex / Omron AOI machine logic used in electronics factories.

Graphics Output
Use `SFML` to render the PCB as a green grid of component slots. As the inspector scans, a highlight box sweeps across each component. Good components turn green, defective ones turn red with a defect-type label (MISSING / BRIDGED / MISALIGNED). A yield counter tracks pass/fail ratio. Inspection speed slider controls scan rate. Final report panel lists all defects with coordinates. Looks exactly like an AOI machine's live screen.

`SFML` PCB grid scan

Machine Vision AOI / PCB IoT Defect Detection

independent objects. SonarArray generates synthetic acoustic ping returns and computes range-bearing maps. DepthController uses a PID loop to hold target depth against buoyancy and current disturbances. ObstacleAvoidance class uses sonar data to steer around objects. TetherManager monitors cable tension and communication latency. Models real BlueROV2 / Oceaneering ROV firmware.

Graphics Output

Use SFML to render a side-view underwater scene — ROV as an icon with animated thruster jets. Sonar ping animates as expanding arc sectors detecting obstacles shown as dark shapes. A top-down sonar map builds progressively as the ROV explores. Depth gauge on the left, thruster power bars on the right. Tether rendered as a cable line from surface to ROV. Ocean floor terrain fills in as sonar maps it. Visually unique — no other project looks like this.

SFML underwater scene

[Underwater IoT](#) [Sonar / SLAM](#) [PID Depth Control](#)